

Technical Document #208: Cybersecurity - Comprehensive Overview

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Penetration testing simulates cyber attacks to identify vulnerabilities. It helps organizations assess their security posture before real attacks occur.

Intrusion detection systems (IDS) monitor network traffic for suspicious activity. They alert administrators of potential security breaches.

Vulnerability management identifies, classifies, and remediates security vulnerabilities. Regular scanning and patching are essential components.

Multi-factor authentication (MFA) requires multiple forms of verification. It significantly reduces the risk of unauthorized access.

Encryption converts plaintext into ciphertext to protect data confidentiality. AES and RSA are widely used encryption algorithms.

Security information and event management (SIEM) systems collect and analyze security logs. They provide real-time visibility into security events.

Autonomous vehicles use a combination of sensors, AI, and mapping to navigate without human intervention. Self-driving technology is rapidly advancing.

Cloud computing has democratized access to powerful computing resources. Startups can now access the same infrastructure as large enterprises.

Digital twins are virtual replicas of physical systems. They enable simulation and optimization without risking real-world resources.

Federated learning trains models across decentralized data sources without sharing raw data. It enables privacy-preserving machine learning.

Edge computing processes data closer to its source, reducing latency and bandwidth usage. It is essential for real-time applications like autonomous vehicles.

Key Takeaways:

- Network segmentation divides networks into smaller segments.
- Vulnerability management identifies, classifies, and remediates security vulnerabilities.
- Penetration testing simulates cyber attacks to identify vulnerabilities.